

App Dev 01 Project Report

1. Student Details

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About Me: I am a student in the IIT Madras BS Degree Program with a strong interest in software development and web application technologies. I enjoy building practical applications that solve real-world problems while improving my understanding of backend development, databases, and user interface design

2. Project Details

Project Title: Trekking Management System

Problem Statement:

To develop a web application for managing trekking activities where administrators manage treks and staff, staff manage assigned treks, and users can browse and book trekking events.

Approach:

The application was built using Flask, SQLite, SQLAlchemy, Bootstrap, and Jinja2. It follows Role-Based Access Control (Admin, Staff, User) and provides trek management, bookings, staff assignment, user management, and profile management through a responsive web interface.

3. AI / LLM Declaration

ChatGPT (GPT-5) was used for learning Flask concepts, debugging errors, UI suggestions, and improving documentation.

The approximate AI usage was **10–15%**. All implementation, integration, testing, and final code were completed manually after understanding the generated suggestions.

4. Technologies and Frameworks Used

<u>Technology</u>	<u>Purpose</u>
Flask	Backend Framework
Flask-SQLAlchemy	ORM
Flask-Login	Authentication
SQLite	Database
Jinja2	HTML Templates
Bootstrap 5	Responsive UI
HTML/CSS	Frontend
Werkzeug	Password Hashing

5. Database Schema / ER Diagram

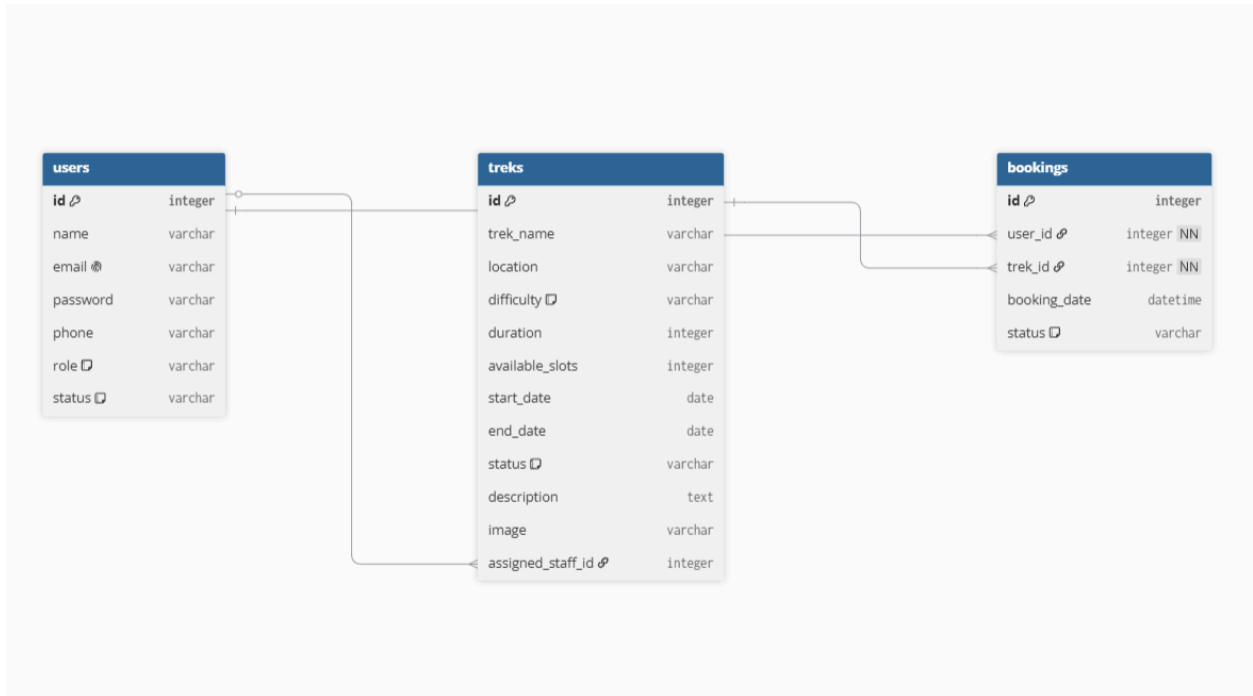
Tables:

1. **User:** id, name, email, password, phone, role, status
2. **Trek:** id, trek_name, location, difficulty, duration, available_slots, start_date, end_date, description, status, image, assigned_staff_id
3. **Booking:** id, user_id, trek_id, booking_date, status

Relationships:

- One User → Many Bookings
- One Trek → Many Bookings
- One Staff → Many Treks

ER diagram image given below:



6. Application Routes

<u>Route</u>	<u>Description</u>
/login	User Login
/register-user	User Registration
/register-staff	Staff Registration
/admin-dashboard	Admin Dashboard
/admin/add-trek	Add Trek
/admin-treks	Manage Treks
/admin/staff	Staff Management
/staff-dashboard	Staff Dashboard
/manage-trek/<id>	Manage Assigned Trek
/participants/<id>	View Participants

/user-dashboard	User Dashboard
/book-trek/<id>	Book Trek
/my-bookings	Booking History
/cancel-booking/<id>	Cancel Booking
/edit-profile	Update Profile

7. Architecture and Features

Architecture

- **app.py** – Application routes
- **models.py** – Database models
- **extensions.py** – Database initialization
- **templates/** – Jinja2 templates
- **static/** – CSS and images

Features

- User & Staff Registration
 - Secure Login System
 - Role-Based Access Control
 - Trek CRUD Operations
 - Trek Image Upload
 - Staff Approval & Assignment
 - Trek Booking & Cancellation
 - Search Functionality
 - Profile Management
 - Responsive Bootstrap UI
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8. Video Presentation

Google Drive Link:

<https://drive.google.com/file/d/1RzuYcsYjMLu5ecB4uxpYDPc68Mi2ojm3/view?usp=sharing>